

Task 2: Deep Reconnaissance, Scanning & Vulnerability Intelligence

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Task: Task 2 – Deep Reconnaissance, Scanning & Vulnerability Intelligence

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GitHub Repository:

https://github.com/pv0101-code/NovaShyld_Task_2

Objective

The objective of this task was to perform advanced reconnaissance on the target machine, identify exposed services, conduct multiple Nmap scans, compare scan results, and analyze the potential attack surface. The task also aimed to perform vulnerability assessment using OpenVAS/Nessus and correlate scan findings into an attack path.

Lab Environment

Component	Details
Host OS	Windows 10
Hypervisor	Oracle VirtualBox
Attacker Machine	Kali Linux Rolling 2025.2
Target Machine	Metasploitable2
Network Mode	Host-Only Adapter
Kali IP	192.168.56.102
Target IP	192.168.56.101

Module 1 – Intelligent Reconnaissance

Host Discovery

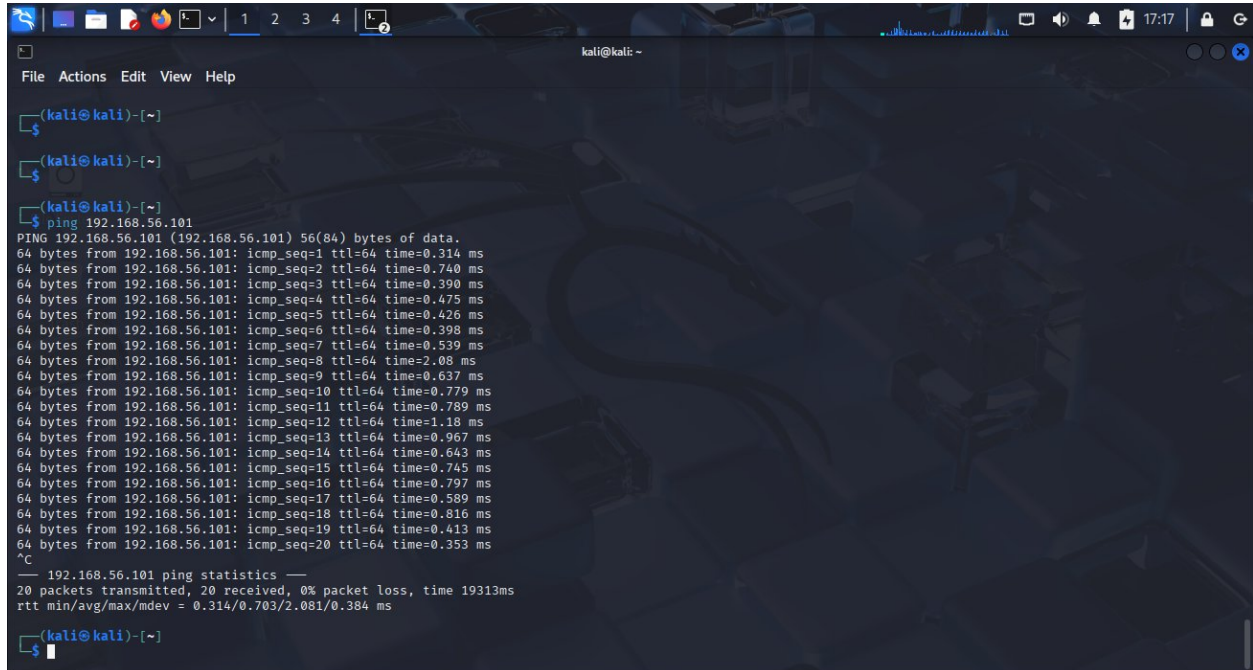
The first step was to verify connectivity between Kali Linux and the Metasploitable2 virtual machine.

Command:

```
ping 192.168.56.101
```

Validation

- Replies received successfully.
- Target confirmed alive.
- No packet loss observed.

A screenshot of a Kali Linux terminal window. The window title is 'kali@kali: ~'. The terminal shows a series of three empty lines, followed by the command 'ping 192.168.56.101'. The output shows 20 successful ping requests with varying response times. At the bottom, a summary line states: '20 packets transmitted, 20 received, 0% packet loss, time 19313ms rtt min/avg/max/mdev = 0.314/0.703/2.081/0.384 ms'.

```
kali@kali: ~  
$  
$  
$  
$ ping 192.168.56.101  
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:  
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.314 ms  
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.740 ms  
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.390 ms  
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=0.475 ms  
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=0.426 ms  
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=0.398 ms  
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=0.539 ms  
64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=2.08 ms  
64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=0.637 ms  
64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=0.779 ms  
64 bytes from 192.168.56.101: icmp_seq=11 ttl=64 time=0.789 ms  
64 bytes from 192.168.56.101: icmp_seq=12 ttl=64 time=1.18 ms  
64 bytes from 192.168.56.101: icmp_seq=13 ttl=64 time=0.967 ms  
64 bytes from 192.168.56.101: icmp_seq=14 ttl=64 time=0.643 ms  
64 bytes from 192.168.56.101: icmp_seq=15 ttl=64 time=0.745 ms  
64 bytes from 192.168.56.101: icmp_seq=16 ttl=64 time=0.797 ms  
64 bytes from 192.168.56.101: icmp_seq=17 ttl=64 time=0.589 ms  
64 bytes from 192.168.56.101: icmp_seq=18 ttl=64 time=0.816 ms  
64 bytes from 192.168.56.101: icmp_seq=19 ttl=64 time=0.413 ms  
64 bytes from 192.168.56.101: icmp_seq=20 ttl=64 time=0.353 ms  
^C  
— 192.168.56.101 ping statistics —  
20 packets transmitted, 20 received, 0% packet loss, time 19313ms  
rtt min/avg/max/mdev = 0.314/0.703/2.081/0.384 ms  
$
```

Host Discovery using Nmap

Command

```
nmap -sn 192.168.56.101
```

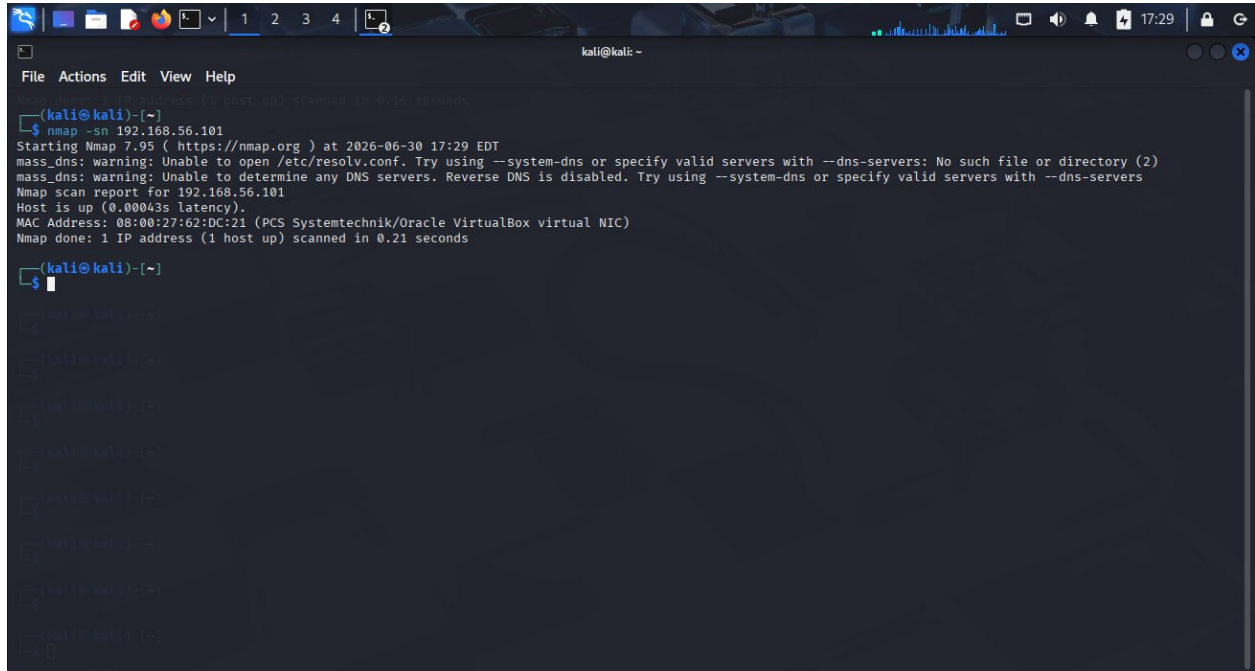
Purpose

The `-sn` option performs host discovery without scanning ports.

Observation

- Host is up.
- MAC address detected.

- Oracle VirtualBox virtual NIC identified.



```
(kali@kali)-[~]
└─$ nmap -sn 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-30 17:29 EDT
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers: No such file or directory (2)
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.101
Host is up (0.00043s latency).
MAC Address: 08:00:27:62:DC:21 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.21 seconds

(kali@kali)-[~]
└─$
```

Banner Grabbing

HTTP

Command

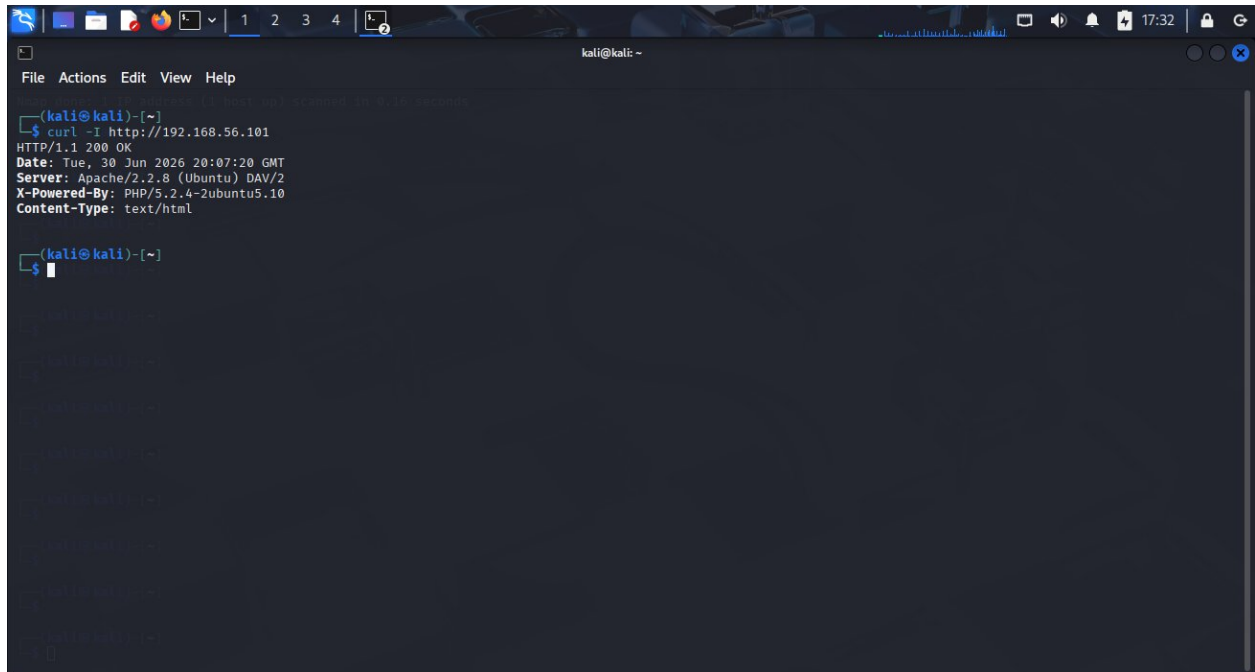
```
curl http://192.168.56.101
```

Purpose:

Retrieve HTTP response headers and webpage information.

Observation:

- Apache web server responded.
- Default Metasploitable web page displayed.



```
(kali@kali)-[~]
└─$ curl -I http://192.168.56.101
HTTP/1.1 200 OK
Date: Tue, 30 Jun 2026 20:07:20 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Type: text/html

(kali@kali)-[~]
└─$
```

The screenshot shows a Kali Linux terminal window. The first command executed is `curl -I http://192.168.56.101`, which returns an HTTP 200 OK response with various headers including Date, Server (Apache/2.2.8), X-Powered-By (PHP/5.2.4), and Content-Type (text/html). Below the terminal output, a list of commands is visible, including `nc 192.168.56.101 21`.

FTP Banner

Command

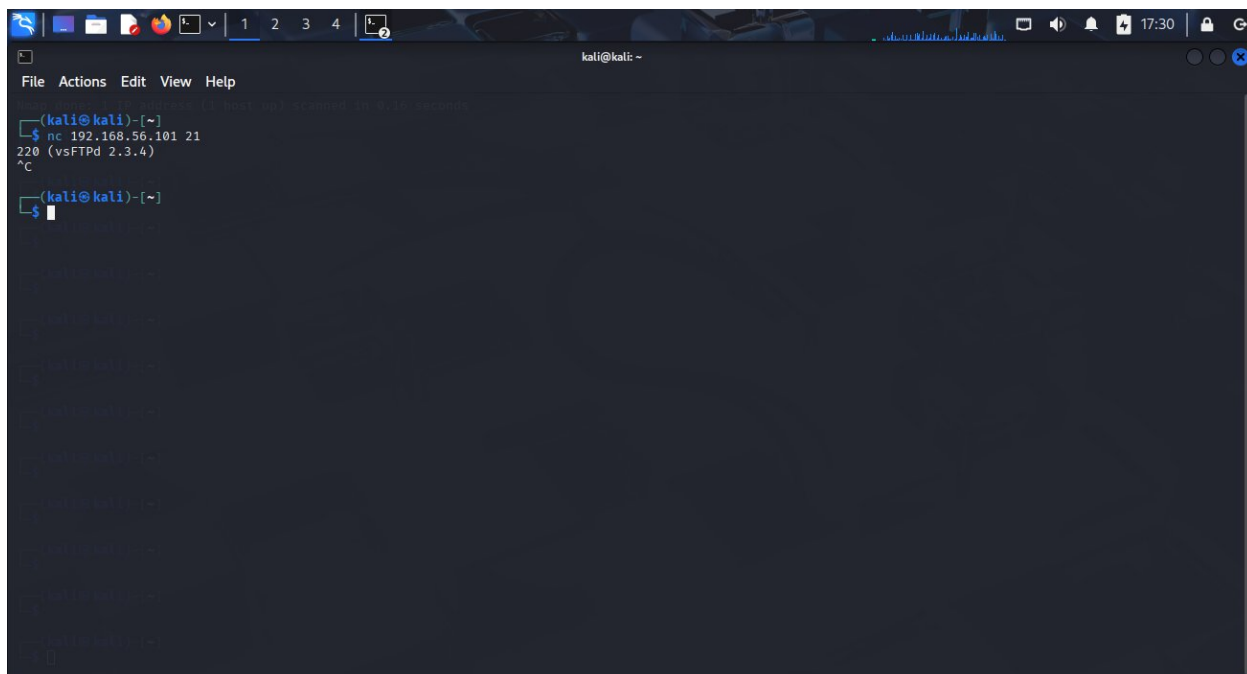
```
nc 192.168.56.101 21
```

Purpose

Retrieve FTP service banner.

Observation

- vsFTPD version detected.
- Service accepts incoming connections.



```
kali@kali: ~  
File Actions Edit View Help  
(Ctrl+Shift+P) Command Palette (Ctrl+Shift+P)  
(kali@kali)~$ nc -l 192.168.56.101 21  
220 (vsFTPD 2.3.4)  
^C  
(kali@kali)~$
```

Services Identified

Port	State	Service
------	-------	---------

21/tcp	open	ftp
--------	------	-----

22/tcp	open	ssh
--------	------	-----

23/tcp	open	telnet
--------	------	--------

25/tcp	open	smtp
--------	------	------

53/tcp	open	domain
--------	------	--------

80/tcp	open	http
--------	------	------

111/tcp open rpcbind

139/tcp open netbios-ssn

445/tcp open microsoft-ds

512/tcp open exec

513/tcp open login

514/tcp open shell

Module 2 – Multi-Layer Scanning

TCP SYN Scan

Command

```
sudo nmap -sS 192.168.56.101
```

Purpose

Performs a stealth SYN scan to identify open TCP ports.

Observation

- Multiple TCP services discovered.
- Several legacy services exposed.

```
kali@kali: ~  
File Actions Edit View Help  
[kali@kali]~$ sudo nmap -sS 192.168.56.101  
[sudo] password for kali:  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-30 17:32 EDT  
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers: No such file or directory (2)  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.101  
Host is up (0.00031s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccrproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
MAC Address: 08:00:27:62:DC:21 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
Nmap done: 1 IP address (1 host up) scanned in 0.50 seconds
```

UDP Scan

Command

```
sudo nmap -sU 192.168.56.101
```

Purpose

Identify UDP services.

Observation

- DNS service detected.
- RPC services identified.

```
kali@kali: ~  
File Actions Edit View Help  
514/tcp open shell  
1099/tcp open rmiregistry  
1524/tcp open ingreslock  
2049/tcp open nfs  
2121/tcp open cproxy-ftp  
3306/tcp open mysql  
5432/tcp open postgresql  
5900/tcp open vnc  
6000/tcp open X11  
6667/tcp open irc  
8009/tcp open ajp13  
8180/tcp open unknown  
MAC Address: 08:00:27:62:DC:21 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
Nmap done: 1 IP address (1 host up) scanned in 0.50 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -sV 192.168.56.101  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-30 17:33 EDT  
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers  
Stats: 0:11:58 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 76.33% done; ETC: 17:49 (0:03:43 remaining)  
Nmap scan report for 192.168.56.101  
Host is up (0.000055 latency).  
Not shown: 952 closed udp ports (port-unreach), 44 open|filtered udp ports (no-response)  
PORT      STATE SERVICE  
53/udp    open  domain  
111/udp   open  rpcbind  
137/udp   open  netbios-ns  
2049/udp  open  nfs  
MAC Address: 08:00:27:62:DC:21 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
Nmap done: 1 IP address (1 host up) scanned in 1022.87 seconds  
  
(kali@kali)-[~]  
$
```

Version Detection

Command

```
sudo nmap -sV 192.168.56.101
```

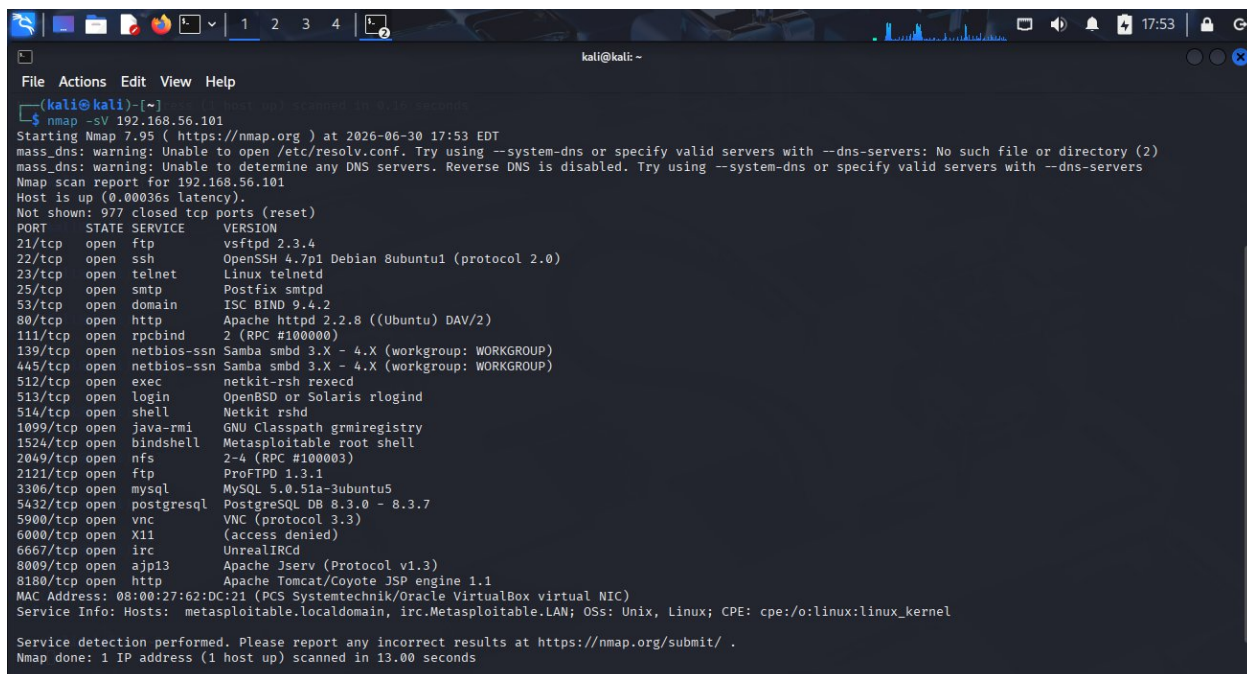
Purpose

Detect service versions.

Observation

Examples:

- vsFTPd 2.3.4
- Apache 2.2.8
- OpenSSH
- Samba



```
kali@kali: ~  
$ nmap -sV 192.168.56.101  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-30 17:53 EDT  
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers: No such file or directory (2)  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.101  
Host is up (0.00036s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind      2 (RPC #100000)  
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec         netkit-rsh rexecd  
513/tcp   open  login        OpenBSD or Solaris rlogind  
514/tcp   open  shell        Netkit rshd  
1099/tcp  open  java-rmi     GNU Classpath grmiregistry  
1524/tcp  open  bindshell    Metasploitable root shell  
2049/tcp  open  nfs          2-4 (RPC #100003)  
2121/tcp  open  ftp          ProFTPD 1.3.1  
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc          VNC (protocol 3.3)  
6000/tcp  open  X11          (access denied)  
6667/tcp  open  irc          UnrealIRCd  
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)  
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1  
MAC Address: 08:00:27:62:DC:21 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 13.00 seconds
```

Operating System Detection

Command

```
sudo nmap -O 192.168.56.101
```

Purpose

Identify the target operating system.

Observation

Nmap correctly identified the target as a Linux-based operating system.

```
kali@kali: ~  
File Actions Edit View Help  
[kali@kali]~$ sudo nmap -O 192.168.56.101  
[sudo] password for kali:  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-30 17:53 EDT  
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.101  
Host is up (0.0010s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rairegistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8000/tcp  open  ajp13  
8180/tcp  open  unknown  
MAC Address: 08:00:27:62:DC:21 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
Device type: general purpose  
Running: Linux 2.6.x  
OS CPE: cpe:/o:linux:linux_kernel:2.6  
OS details: Linux 2.6.9 - 2.6.33  
Network Distance: 1 hop  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 2.39 seconds  
[kali@kali]~$
```

Comparison of Scan Results

Scan	Information Obtained
Host Discovery	Verified target availability
TCP SYN	Open TCP ports
UDP	UDP services
Version Detection	Service versions
OS Detection	Operating System
Aggressive Scan	Combined information

Module 3 – Vulnerability Intelligence

Objective

Perform vulnerability assessment using OpenVAS or Nessus to identify critical and high-risk vulnerabilities.

Current Status

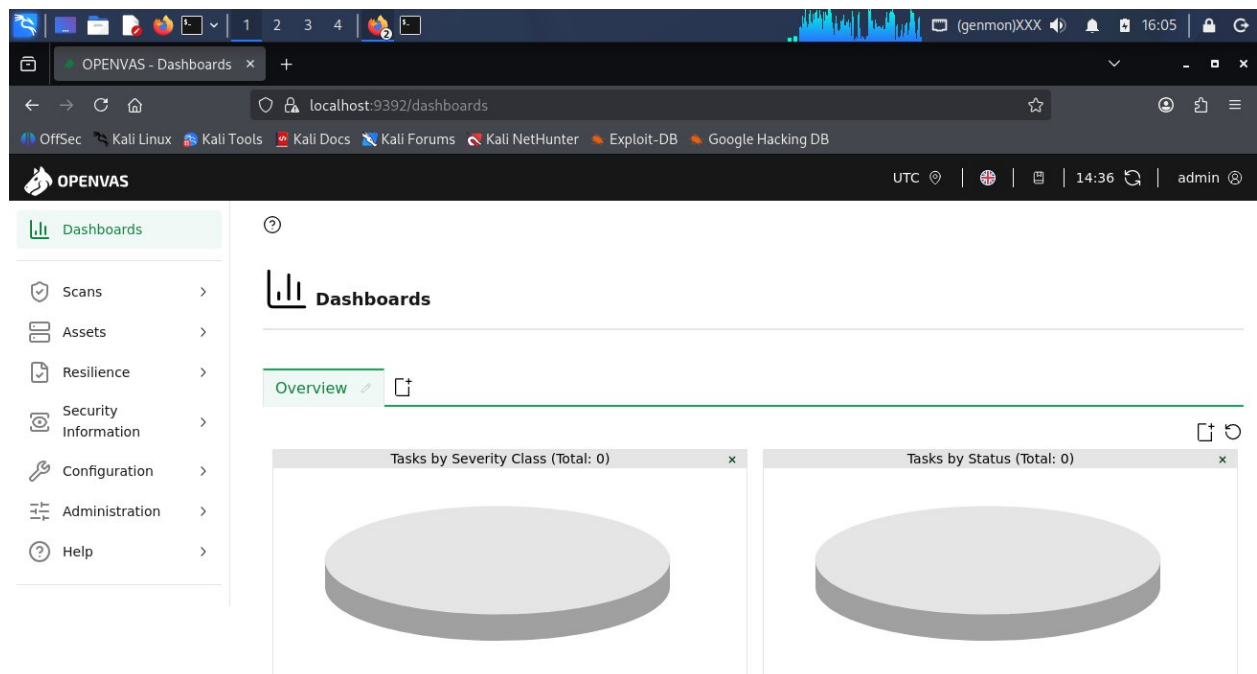
OpenVAS (Greenbone Vulnerability Manager) was successfully installed and configured. Feed synchronization and database initialization were initiated successfully. However, during the initial vulnerability feed import, the `gvmd` service was repeatedly terminated by the Linux Out-Of-Memory (OOM) killer due to limited virtual machine memory, preventing completion of the initial database build and scan configuration import.

Evidence Collected

- Successful installation of GVM components
- Successful feed synchronization initiation
- `gvm-check-setup` completed successfully
- `gvmd` service active
- Kernel logs confirmed OOM termination during SCAP database import

Planned Resolution

The vulnerability assessment will be completed after increasing the virtual machine memory allocation and allowing the initial feed synchronization to complete successfully.

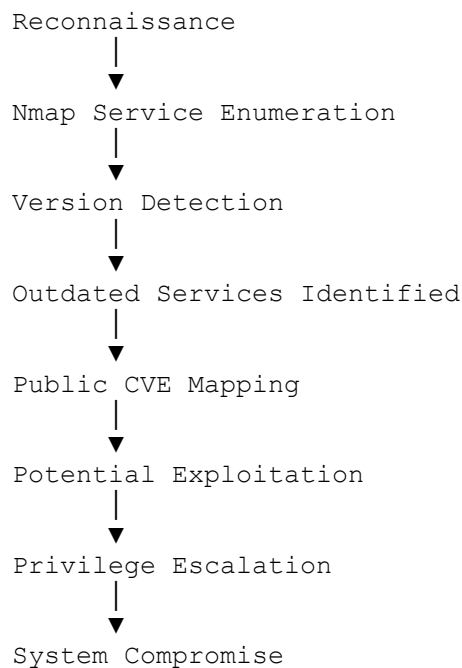


Module 4 – Correlation & Attack Path Mapping

Although the automated vulnerability scan could not be completed, the exposed services identified during reconnaissance indicate several realistic attack paths.

Service	Version	Possible Weakness	Potential Exploit
FTP	vsFTPD 2.3.4	Known backdoor vulnerability	Remote shell access
Telnet	Legacy service	Plaintext authentication	Credential interception
Apache	2.2.x	Outdated web server	Known web exploits
Samba	Older version	Remote code execution vulnerabilities	SMB exploitation
Tomcat	8180	Weak/default credentials	Manager application compromise

Prioritized Attack Path



Conclusion

The reconnaissance and scanning phases successfully identified multiple exposed services and legacy software running on the Metasploitable2 target. These findings reveal a broad attack surface and several potential exploitation paths. While the automated vulnerability assessment could not be finalized due to a resource limitation during the initial GVM feed import, the environment was successfully configured and verified, and the remaining assessment can be completed after allocating additional memory to the virtual machine.